

Group Assignment 1 Instructions

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1 Problem description

Your team has been hired to help with a data migration to a Lakehouse. The goal is to design the bronze, silver and gold layers, and to build the data processing pipeline that populates data across them. The team also needs to help the customer become more data-driven by answering a set of specific business questions.

Customer: TPC-H — a decision-support benchmark modelling a wholesale supplier.

This is the first of two project phases. To compete for the second one, your team has to complete this one.

2 Benefits

Each team member can earn up to **7 points**. To earn the maximum, submit the presentation before the due date and present it in class.

3 List of deliverables

3.1 Lakehouse layers

- **Bronze layer** — store the data as is, no changes; add metadata only if needed.
- **Silver layer** — should be in 3NF, with enforced data quality.
- **Gold layer** — should be designed to answer the business questions (see [Customer profiles](#), and pick the one your team chose in the chat).

3.2 Repository

A public repo with a working solution and the presentation:

- use `uv`
- code (Python or SQL)
- presentation (or a link to it in the README)
- a meaningful README

Ideally, the scripts should be easily portable to another workspace:

- think about naming
- assume that your team only has access to pre-production data

3.3 Databricks

Feel free to use:

- notebooks
- Python files
- Databricks Jobs & Pipelines (if you find it relevant)

3.4 Presentation

- 5–7 minutes to explain your solution
- team member responsibilities
- link to the repo
- the challenges you faced
- a short demo, including how your team decided to perform validation
- a screenshot of the ER diagram of the silver tables
- answers to the business questions — come up with the code and a visualisation (use a notebook and/or a Databricks dashboard)

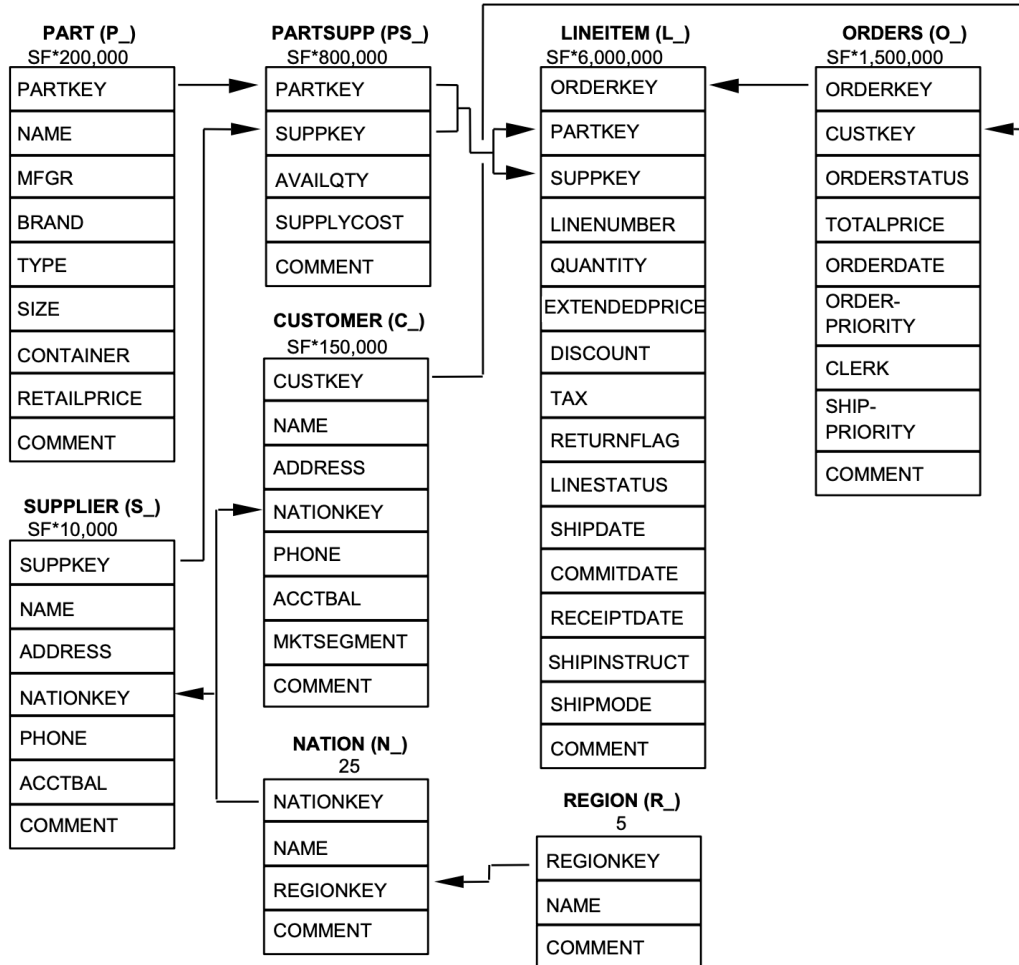
4 Customer data

TPC-H data ships with every Databricks workspace as `samples.tpch` — 8 tables, no download, no setup.

Tip: a good starting point for understanding the data is its README:

```
with open("/dbfs/databricks-datasets/tpch/README.md", "r") as f:
    content = f.read()

displayHTML(f"<pre>{content}</pre>")
```



5 Customer profiles

Each customer profile contains:

- a short description
- a set of typical questions the customer needs your help with
- validation rules
- monitoring — the metrics the customer wants to watch over time
- alerting to bring customer's attention

While questions you should be able to answer using gold layer, validation rules can use all layers to prove the data can be trusted.

5.1 Marketing

Marketing team works on customer acquisition and retention. The team utilizes metrics such as activation rate (the share of customers with at least one order) by segment, repeat purchase rate, and the dynamics of new customers over time.

Questions

1. What is the activation rate (share of customers with at least one order) for the building segment?
2. How many new customers (by their first order date) appeared in this quarter (1996-Q1)?
3. Which segment has the highest repeat purchase rate (share of customers with more than one order)?
4. Group customers by the month of their first order (cohort). For each 1996 cohort, what share of customers placed another order within the next three months?

Validation rules

- Every customer has a valid, non-null market segment.
- Every order has a valid customer reference.
- Customers with no orders survive into silver and appear in gold as zero-activity, rather than being silently dropped by an inner join.

Monitoring

Track how your key metrics move over time: activation rate by segment, and the count of new customers per period. Optional: decide what a meaningful drop would look like and set an alert for it.

5.2 Logistics

Logistics team is responsible for the timeliness and consistency of deliveries. The main table is `lineitem`: checking dates (shipping, confirmation, receipt), analyzing delays, and evaluating the performance of different shipping modes (`shipmode`).

Questions

1. For each ship mode, report the median and the p90 transit time (ship date to receipt date). Which mode is fastest, and which is most predictable? Explain why those are not the same question.
2. An order is fully on time only if every one of its line items arrived by its commit date. What share of orders is fully on time, and how does that compare with the share of line items on time? Explain the gap.
3. For each ship mode, what share of line items was received after the commit date? Which mode performs worst?

4. Are urgent-priority orders fulfilled faster than others? Support your answer with numbers.

Validation rules

- dates on each line item make sense relative to each other and to the parent order (a shipment can't leave before the order was placed, can't be received before it was shipped)
- Categorical fields (ship mode, return flag, line status) contain only values you'd expect — build the allowed list by profiling the source and explain how you got it.
- Every line item belongs to an existing order, and no order has zero line items.

Monitoring

Track the delay rate (share of line items received after the committed date) over time. Optional: alert on a meaningful increase.

5.3 Brand Manager

The team analyzes the effectiveness of brands and manufacturers: revenue by brand, brand market share within a product category, and the difference between list price and cost price.

Questions

1. What is the total revenue of Brand#32, and what is its share within the product category it belongs to?
2. What is the average margin (list price minus supply cost) for products of Brand#32?
3. Which brand has the largest share of sales within its category, and is that share stable over time?
4. Find products whose actual sale price differs from the list price by more than 15%. How many such products are there, and are they concentrated in a particular brand or category?

Validation rules

Prove that brand-level analysis can be trusted:

- every part has a non-empty brand and manufacturer
- revenue/margin calculations are consistent (no double-counting across suppliers of the same part)

Monitoring

Track brand revenue and market share within its product category over time. Optional: alert on a meaningful shift in a brand's share (gain or loss).

5.4 Finance

The team is responsible for the financial accuracy of data: reconciling order totals with the sum of individual line items, verifying business rules (discount ranges, positive values for prices and quantities), and comparing monetary columns between the source and each data layer.

Questions

1. Which 10 customers generate the most net revenue, and what share of total revenue do they account for?
2. Report 1997 revenue three ways — gross extended price, net of discount, and net of discount and tax. How far apart are the three figures, and which one should the business report? Defend the choice in one paragraph.
3. Which market segment has the highest average order value, and does that change year over year?
4. Which 5 suppliers generate the most revenue, and what is their average discount?

Validation rules

- Every order total matches the sum of its line items (use a fixed tolerance of \$0.01).
- Discount and tax fall inside their allowed range — derive the bounds by profiling the source and justify them in the readme.
- Quantity, extended price and retail price are strictly positive.
- Monetary totals agree across bronze, silver and gold: compute the same net revenue figure per layer and reconcile them.

Monitoring

Track the count of mismatched orders over time. Optional: alert if a new mismatch appears.

5.5 Strategist / CTO

The team looks at the company as a whole: revenue dynamics, number of active customers, and top suppliers and products by contribution to overall results. They are responsible for business-oriented alert rules (for example, a sharp drop in revenue).

Questions

1. What is total revenue for quarter 1996-Q1, and how does it compare to the previous quarter?
2. How many active customers (at least one order in the last year of data) were there as of 1998-08-02?
3. If revenue dropped 20% week over week, which alert rule would catch it? Show the query and demonstrate it against artificially reduced data.
4. Which 3 metrics best represent “company health” at a glance, and why these three?

Validation rules

- Every headline number has one written definition in the readme, and every query in your solution uses it — e.g. revenue is net of discount; an active customer is one with at least one order in the trailing 12 months.
- Any period-over-period comparison uses a consistent time boundary (no partial periods compared to full ones)

Monitoring

Build a small set of headline metrics (revenue trend, active customers, top contributors)
Optional: design alerting for when the CTO needs to pay attention.

5.6 Regional Manager

Regional Manager analyzes sales through a geographical hierarchy: top products by region, ranking regions by revenue, and products with limited geographical coverage.

Questions

1. What are the top 5 products by revenue in Asia?
2. How many customers are in India, and what is their average account balance?
3. Which region has the highest total revenue, and what share of sales happens between a customer and a supplier in different regions?
4. Are there products sold in only one or two regions? What might that indicate?

Validation rules

- Every customer and supplier resolves to a valid nation, and every nation to a valid region.
- Revenue attributed to a region sums correctly and doesn't double-count across regions.

Monitoring

Track revenue by region over time. Optional: alert on a meaningful shift in a region's rank or volume.

5.7 Procurement

The Procurement team is responsible for where the company buys: what it pays suppliers, how exposed it is to any single one, and which suppliers are worth keeping. The main tables are supplier and partsupp — note that a part is linked to a supplier by both keys together, not by one of them.

Questions

1. Which 10 suppliers carry the largest inventory value (available quantity $\times$ supply cost), and what share of the total do those 10 hold?
2. For the parts of Brand#32, how much does supply cost vary between the suppliers that offer the same part — cheapest against most expensive? Are we actually ordering from the cheapest one?
3. For how many parts did we order from a single supplier only, even though partsupp lists others who could have supplied it? What is the risk, and is it concentrated anywhere?
4. Some suppliers have customer complaints recorded against them, but not in a column you can filter on — the signal is in free text. Find those suppliers, and work out what share of our spend goes through them.

Validation rules

- Every line item's (part, supplier) pair exists in partsupp — this is a two-column foreign key, so explain how you enforced it.
- Every supplier resolves to a valid nation, and every nation to a valid region.
- Supply cost is positive and never above the part's retail price.

Monitoring

Track spend concentration (the share held by the top 10 suppliers) and average supply cost by region over time. Optional: alert when a single supplier's share of spend crosses a threshold you choose.